

K2-29b

R-2114

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_220225 · created 2026-08-11T00:33:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459635.6598 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.05
Transit depth (Rp/Rs)^2	2.95 +/- 1.71 [%]
Orbital Inclination (inc)	89.64 +/- 2.43
Airmass coefficient 1 (a1)	0.99 +/- 0.19
Airmass coefficient 2 (a2)	-0.005 +/- 0.026
Scatter in the residuals of the lightcurve fit is	19.17 %
Best Comparison Star	#2 - [417, 165]
Optimal Aperture	4.24
Optimal Annulus	12.39
Transit Duration (day)	0.0677 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.05 vs 0.1419 (deviation 0.3σ)
Duration fit vs published	0.0677 d vs 0.0925 d (deviation 1.26σ)
Mid-time O–C	11.8 min
Depth SNR	1.7
Residual scatter	19.17 %

Light curve

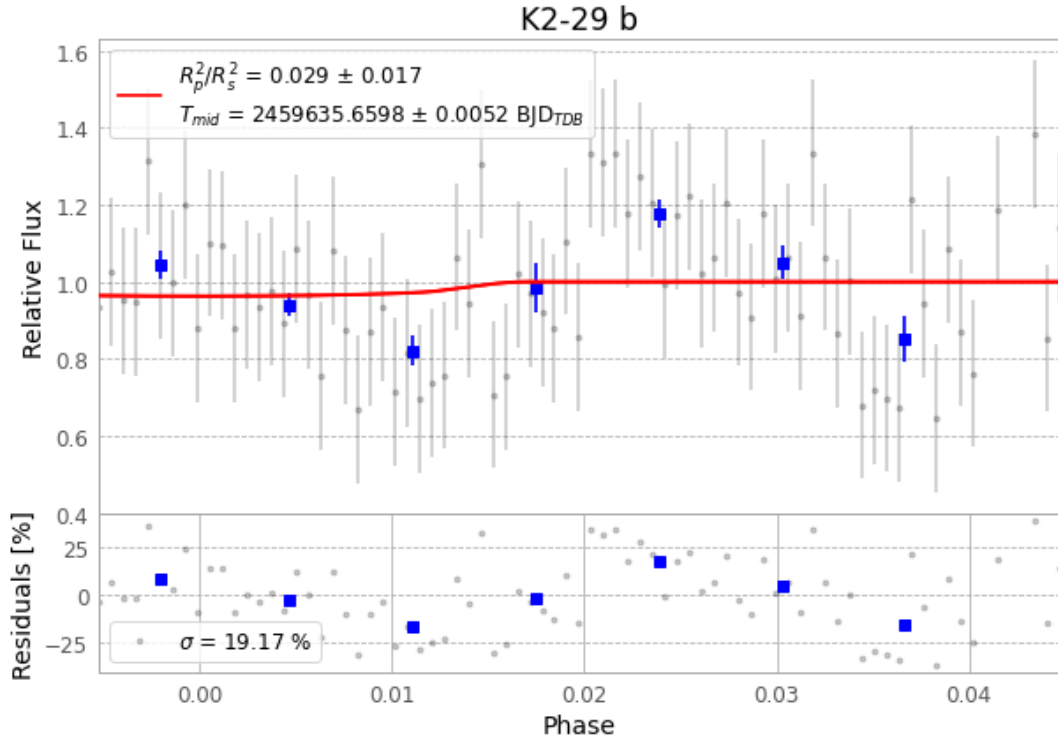


Figure 1 — FinalLightCurve_K2-29 b_2022-02-24.png (SHA-256 b952028289730304...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 412], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e5502f3c224183...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

79 FITS frames (52 MB), K2-29220225032415.FITS → K2-29220225072412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-220225.log (d67c38711ae4...), FinalLightCurve_K2-29 b_2022-02-24.png (b95202828973...), FinalLightCurve_K2-29 b_2022-02-24.csv (c6c1820ae95f...)

Compute resources

Wall time 293.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:33Z.